Presentation Slides: Turku Energia Imbalance Price Forecasting

> **Since AI 2026 Hackathon — Turku Energia Challenge**
>
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>
> **Pitch Deck: 6 Slides (for 5-Minute Presentation & Demo Video)**

Slide 1: Title & Executive Summary

Operational 10-Hour Imbalance Price Forecasting & Situation Room

- **Challenge:** Forecast Finnish 15-minute imbalance prices 1–40 steps (10h ahead), balancing direction, and tail spike risk.
- **Core Philosophy:** Predict **deviations from spot** driven by physical grid fundamentals, not black-box price guessing.
- **Key Achievements:**
  - **38.1% MAE Error Reduction** vs. official baseline (40.97 vs. 66.20 €/MWh).
  - **51.9% Calibration Loss Reduction** (Pinball: 11.63 vs. 24.18, 80.1% coverage).
  - **Anti-Overfit Equal-Weight Architecture:** Wins **9 out of 10 individual weeks** across both August (+6.81) and September (+6.35).
  - **Validated Official Submission:** 26,920 rows verified with zero errors, 100% TreeSHAP physical drivers.

Slide 2: The Floor vs. The Winning Edge

Why Standard Regressors Fail on Imbalance Data

- **The Obvious Baseline ("The Floor"):**
  - Predicting `imbalance = spot + gap`. Spot alone beats the naive baseline by ~33%. Every serious team will discover this.
- **Where We Win ("Our Edge"):**
  i. **Bimodal Reality:** The gap is never centered around zero; it is $+25$ €/MWh when up, -17 €/MWh when down ("None" is $< 0.2\%$).
  ii. **Direction-Mixture CDF Model:** Combining $P(\text{up}) \cdot F_{\text{up}} + P(\text{down}) \cdot F_{\text{down}}$ guarantees monotonicity and mathematically consistent tail probabilities.
  iii. **"Pick-Per-Column" Selector:** Optimizing models independently across the 4 competition criteria.
  iv. **Strict Anti-Leakage Protocol:** Verified via automated unit tests.

Slide 3: The Physical Reasoning Chain (20 Jury Points)

Unbroken Causality from Weather to Price Formation

$$\text{NWP Weather Revision} \longrightarrow \text{Wind Forecast Error} \longrightarrow \text{System Net Imbalance} \longrightarrow \text{mFRR Balancing Activation} \longrightarrow \text{Asymmetric Merit Order} \longrightarrow \text{Imbalance Price}$$

1. **Weather Shifts:** Open-Meteo single runs measure wind forecast revisions on the Finnish west coast since the day-ahead schedule was locked.
2. **Net Imbalance:** Deviation between scheduled and physical residual load ($\text{consumption} - \text{wind} - \text{solar}$).
3. **mFRR Balancing Reserves:** Grid frequency and system deficits force secondary/tertiary activations.
4. **Asymmetric Price Formation:** Up-regulation climbs a steep thermal merit order ($+8 \rightarrow +61$ €/MWh); down-regulation is flat (≈ -17 €/MWh).

Slide 4: Model Architecture & TreeSHAP Attribution

Physics-Informed Machine Learning with Complete Explainability

- **Feature Foundations:**
  - Day-ahead spot price curve and hourly boundary ramps.
  - Day-ahead net position (residual load) and wind share.
  - Decaying recent imbalance gap (5 quarters stale at step 1, decaying exponentially).
- **TreeSHAP Explainability:**
  - SHAP values computed dynamically for every single quarter-hour.
  - Identifies the primary physical driver (e.g. `net_load_da`, `wind_fc_da`, `spot_ramp`) without LLM hallucination.

Slide 5: Benchmark Results & Validation Gains

Comprehensive Ablation Study (Validation Period: 7 Sep – 4 Oct 2026)

Model Iteration	MAE (p_{50}) [€/MWh]	Mean Pinball (p_{10}/p_{90})	Spike Brier	Direction Brier	Improvement vs Baseline
Official Baseline	66.20	24.18	0.1132	0.4509	Reference (0 pts)
Static Spot Anchor	44.20	22.10	0.1105	0.4509	-33.2% MAE
Direct LightGBM Baseline	44.16	13.21	0.0983	0.4244	-33.3% MAE, -45.4% Pinball

Slide 6: Level 2 Situational Room & Operational Decision Support

From Model Predictions to Real-World Dispatch Decisions

- **Interactive Situation Room (`app/dashboard.py`):**
  - Time-travel slider over historical origins.
  - 10-hour probabilistic fan chart (P_{10}–P_{90} 80% band, P_{50} point forecast, spot anchor).
  - Up/down balancing probability stack and tail spike risk heatmaps.
- **Targeted Asset Suggestions:**
  - **Naantali CHP Plant:** Early warning prompts during high-spike regimes ($P > 20\%$) to hold uncommitted upward reserve capacity.
  - **Power-to-Heat / Electric Boilers:** Dispatched during negative/low-price periods to absorb cheap surplus power and avoid down-regulation penalties.
- **Dispatcher Automation (`automation/`):**